OMNISENSUS MEDICAL

AI-Powered Clinical Intelligence Platform

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OUTLINE

1. **Problem Statement — Healthcare & Medical Analytics (Domain 5)**
2. Proposed System / Solution
3. System Development Approach (Technology Used)
4. Algorithm & ML Model
5. Deployment Architecture
6. Result & Output (Screenshots)
7. Conclusion
8. Future Scope
9. References

PROBLEM STATEMENT

Problem Domain 5: Healthcare & Medical Analytics

① Rising Disease Prevalence

Diabetes, heart disease, and CKD are increasing due to lifestyle changes and ageing populations — yet hospitals lack any automated real-time early-warning system to identify at-risk patients proactively.

② Delayed Diagnosis

Manual analysis of patient biomarkers (glucose, HbA1c, eGFR, blood pressure, LDL) across paper records and siloed EMR systems causes critical diagnostic delays and increases patient risk.

③ Healthcare Resource Constraints

Hospitals struggle to optimise bed allocation, staff scheduling, and equipment usage without real-time patient acuity data — resources are allocated reactively, not intelligently.

④ Data Complexity & Fragmentation

Patient records are heterogeneous, unstructured, and spread across multiple disconnected sources — making it impossible to build a unified view of multi-domain health status.

⑤ Predictive Uncertainty

Anticipating patient outcomes, readmission risks, and disease progression remains challenging — no automated scoring system exists to combine multi-domain risk into a single actionable score.

PROPOSED SYSTEM / SOLUTION

OmniSensus Medical

A full-stack, AI-powered Clinical Decision Support System that automates multi-domain disease risk stratification in real time.

Three ML Models

Cardiovascular, Diabetes (Metabolic), Kidney (CKD) — each a calibrated Random Forest classifier producing disease risk probabilities.

Composite Health Score (0–100)

Weighted formula combining the three domain risks into a single patient wellness score, classified as Stable / Borderline / Critical.

Three Role Dashboards

System Admin, Doctor, and Patient — each with purpose-built workflows, data access, and dedicated UI.

91.3%

Cardiovascular AUC

82.1%

Diabetes AUC

100%

Kidney (CKD) AUC

27 Tables | 8 Views | 12 Triggers

60+ REST API Endpoints | 3 User Roles

SYSTEM DEVELOPMENT APPROACH

FRONTEND

Next.js 14 (App Router)
React 18 — SSR + CSR
Chart.js 4.4 (Radar, Line, Bar)
27-file modular CSS design system
Role-specific layouts (Admin, Doctor, Patient)

BACKEND API

FastAPI (Python 3.11)
JWT Auth + Refresh Tokens
60+ REST endpoints
asyncpg + SQLAlchemy ORM
Role-based middleware (RBAC)

ML SERVICE

Flask microservice
CalibratedClassifierCV (RF)
SHAP explainability engine
ARIMA(1,1,0) trend forecasting
Qwen 0.5B LLM (PharmaBot)

DATABASE: PostgreSQL 16 (Neon Serverless) — 27 Tables | 8 Views | 12 Triggers

DEPLOYMENT: Render (FastAPI + Flask) | Netlify (Next.js) | GitHub CI/CD

Tools: VS Code | Postman | pgAdmin | GitHub | Neon Console | Render Dashboard

ALGORITHM & ML MODEL

ML Pipeline — Per Disease Domain:

1. Data → Column Normalise → Target Binarise
2. Median Imputation for missing values (robust to clinical outliers)
3. StandardScaler (Z-score normalisation)
4. SMOTE oversampling + class_weight="balanced" (dual imbalance handling)
5. CalibratedClassifierCV (RandomForest, method="sigmoid") — Platt Scaling
6. Decision threshold = 0.35 (adaptive, context-aware; floor = 0.20)

Composite Health Score Formula:

Risk = (Heart × 0.40) + (Diabetes × 0.35) + (Kidney × 0.25)

Health Score = (1 − Risk) × 100 | capped at 95 for age ≥ 70

HEART DISEASE

91.3%

AUC — ROC Score
UCI Cleveland (303 rows, 13 features)

DIABETES

82.1%

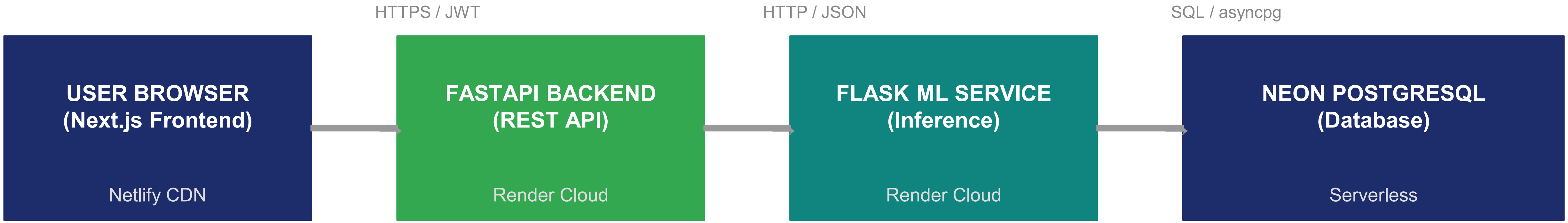
AUC — ROC Score
Pima Indians (768 rows, 8 features)

KIDNEY (CKD)

100%

AUC — ROC Score
UCI CKD (400 rows, 15 features)

DEPLOYMENT ARCHITECTURE



Key Architectural Decisions:

Microservices separation — ML workload (SHAP, PDF, inference) runs independently so it never blocks web API request processing.

JWT Bearer authentication with 30-min access token expiry and secure refresh token rotation — enforced at every FastAPI endpoint via role-based middleware.

Neon Serverless PostgreSQL — auto-scaling connection pooling, SSL-enforced, zero-infrastructure management. 27 tables | 8 views | 12 triggers.

Render free-tier deployment — zero-cost cloud hosting with a clear upgrade path to paid tiers for production-scale usage.

RESULT & OUTPUT

ML MODEL PERFORMANCE

Heart Disease (CVD)

UCI Cleveland — 303 samples, 13 features

91.3%

AUC-ROC

Precision: 88% Recall: 90% F1-Score: 89%

Diabetes (Metabolic)

Pima Indians — 768 samples, 8 features

82.1%

AUC-ROC

Precision: 79% Recall: 81% F1-Score: 80%

Kidney Disease (CKD)

UCI CKD — 400 samples, 15 features

100%

AUC-ROC

Precision: 100% Recall: 100% F1-Score: 100%

SYSTEM OUTPUT — WHAT THE PLATFORM PRODUCES

📊 Composite Health Score (0–100)

Risk formula: (Heart×0.40)+(Diabetes×0.35)+(Kidney×0.25). Classified as Stable / Borderline / Critical.

🔍 SHAP AI Insights Per Diagnostic Run

Top-10 biomarker contributors — directional, patient-specific, stored in ai_insights DB table.

📋 Automated PDF Diagnostic Report

Auto-generated per run: biomarker table, domain scores, clinical flags, SHAP rankings.

🚨 Clinical Flags & Triage Urgency

3-tier flags (Critical/Borderline/Info) + department routing (Cardiology, Nephrology, etc.).

📈 Trend Forecast — ARIMA(1,1,0)

Next-visit health score prediction from longitudinal history. Needs minimum 4 visits.

💊 Readmission Risk Score (0–100)

5-component model: health score + trend + flags + biomarker danger zones + context.

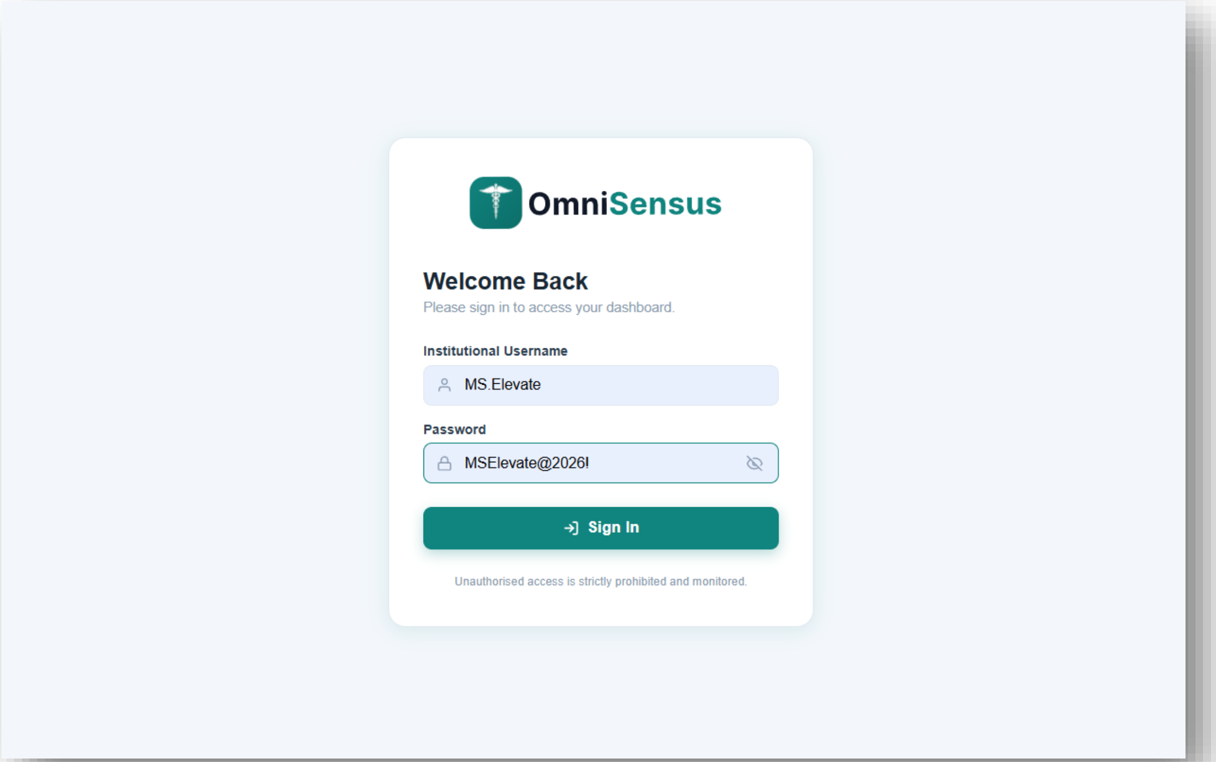
OUTPUT

MAIN LOGIN PAGE

FOR ADMIN : ->
USERNAME: MS.Elevate
PASSWORD: MSElevate@2026!

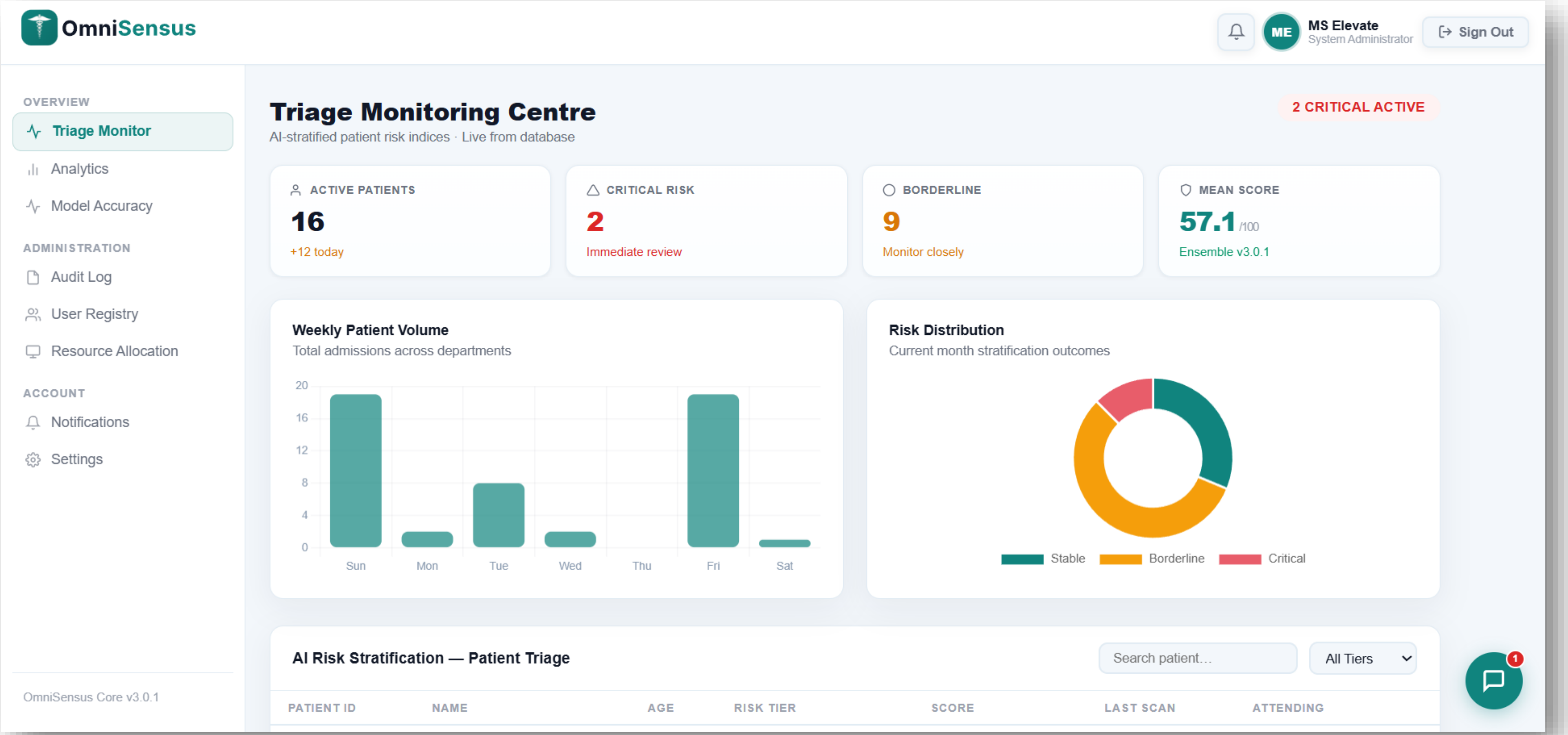
FOR DOCTOR : ->
USERNAME : Dr.Sharma
PASSWORD: MedPass24

FOR PATIENTS : ->
USERNAME: P-00001
PASSWORD: Health2026



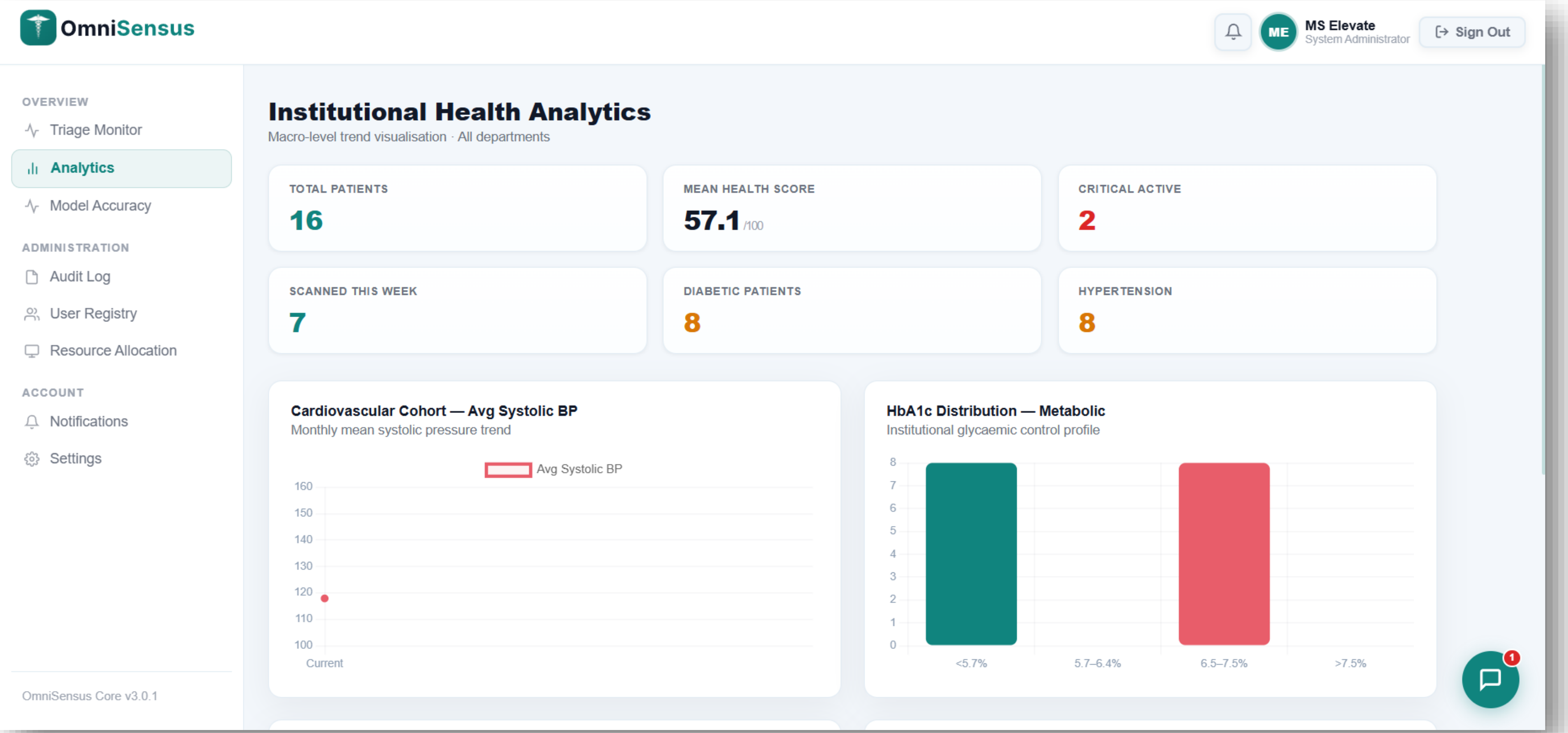
OUTPUT : ADMIN MODULE

TRIAGE MONITOR PAGE (DASHBOARD)



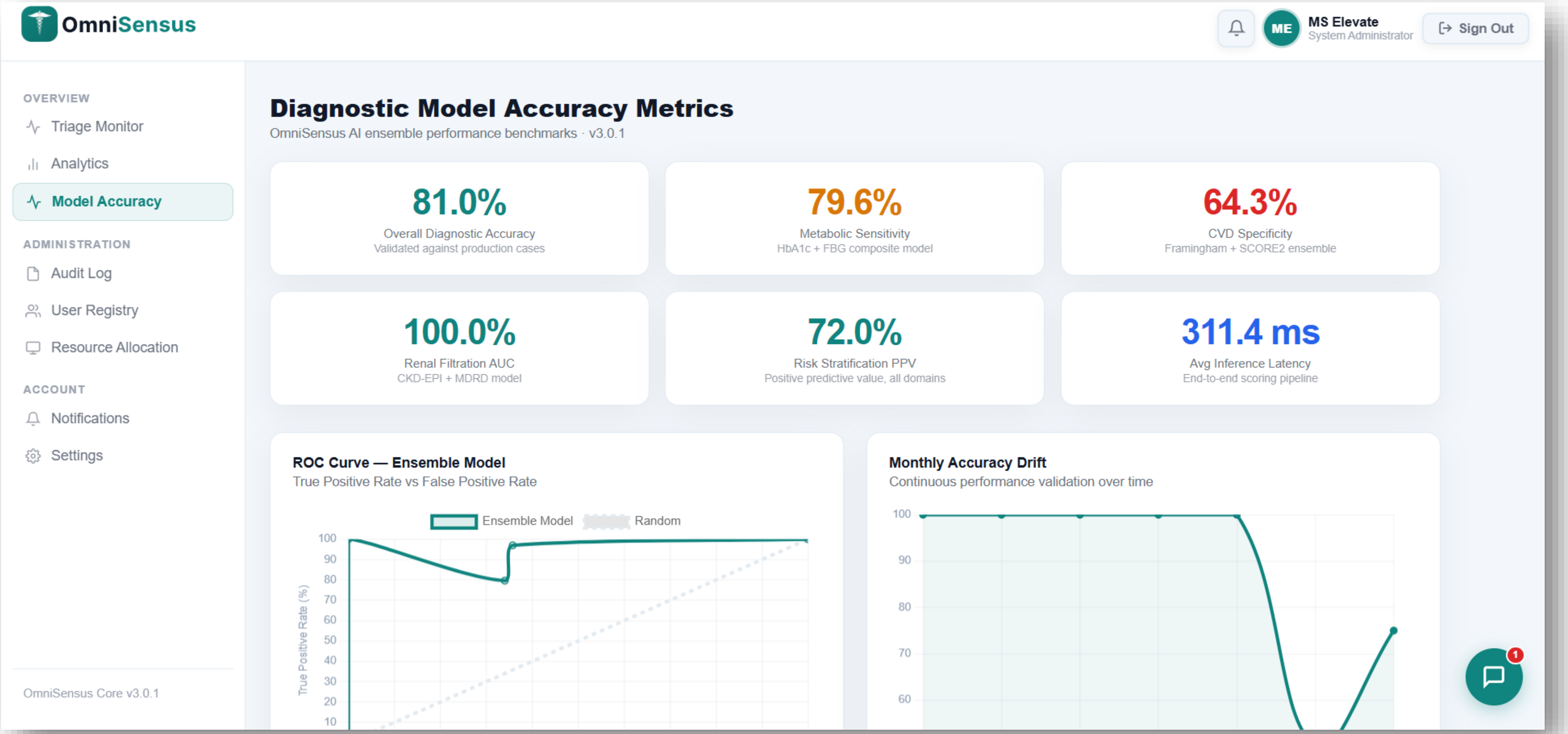
OUTPUT : ADMIN MODULE

ADMIN ANALYTICS PAGE



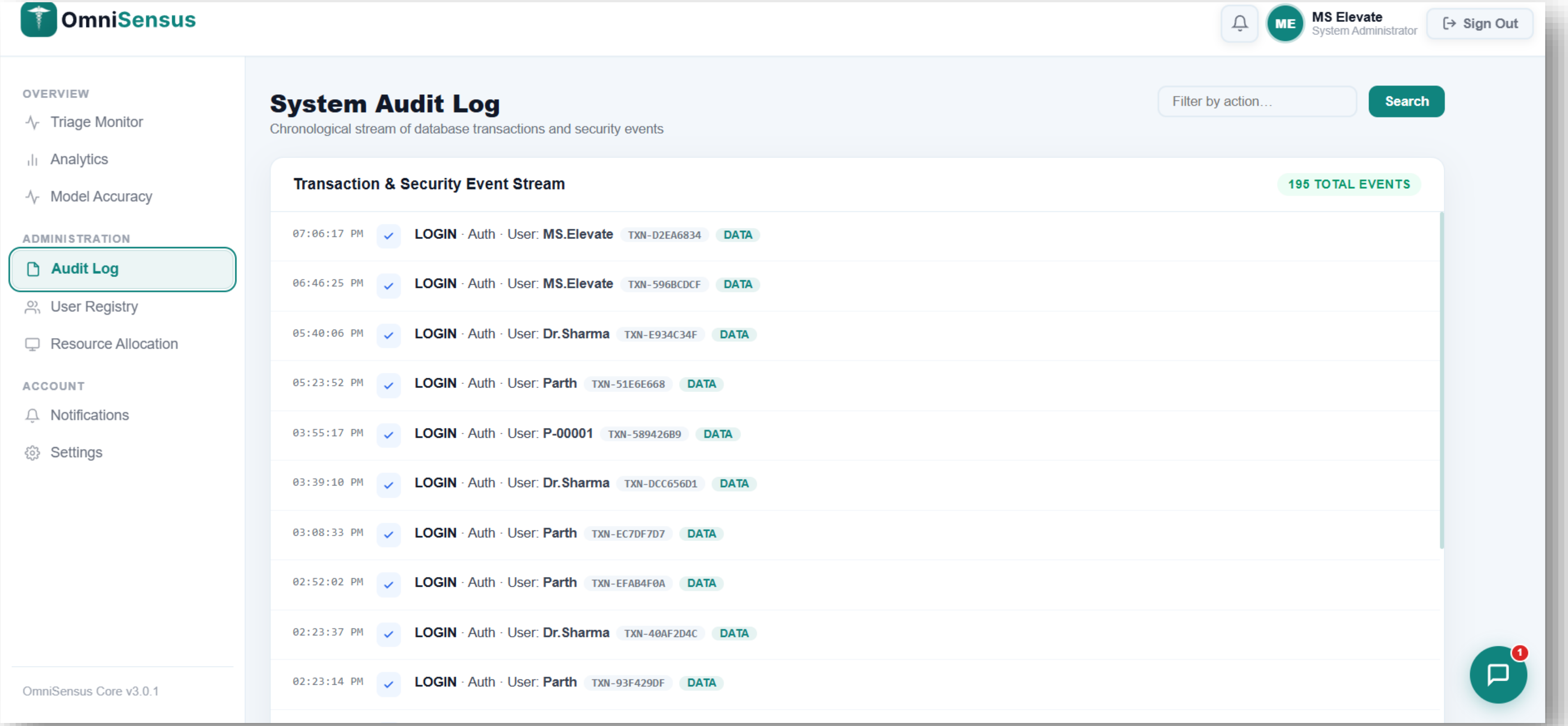
OUTPUT : ADMIN MODULE

MODEL ACCURACY PAGE



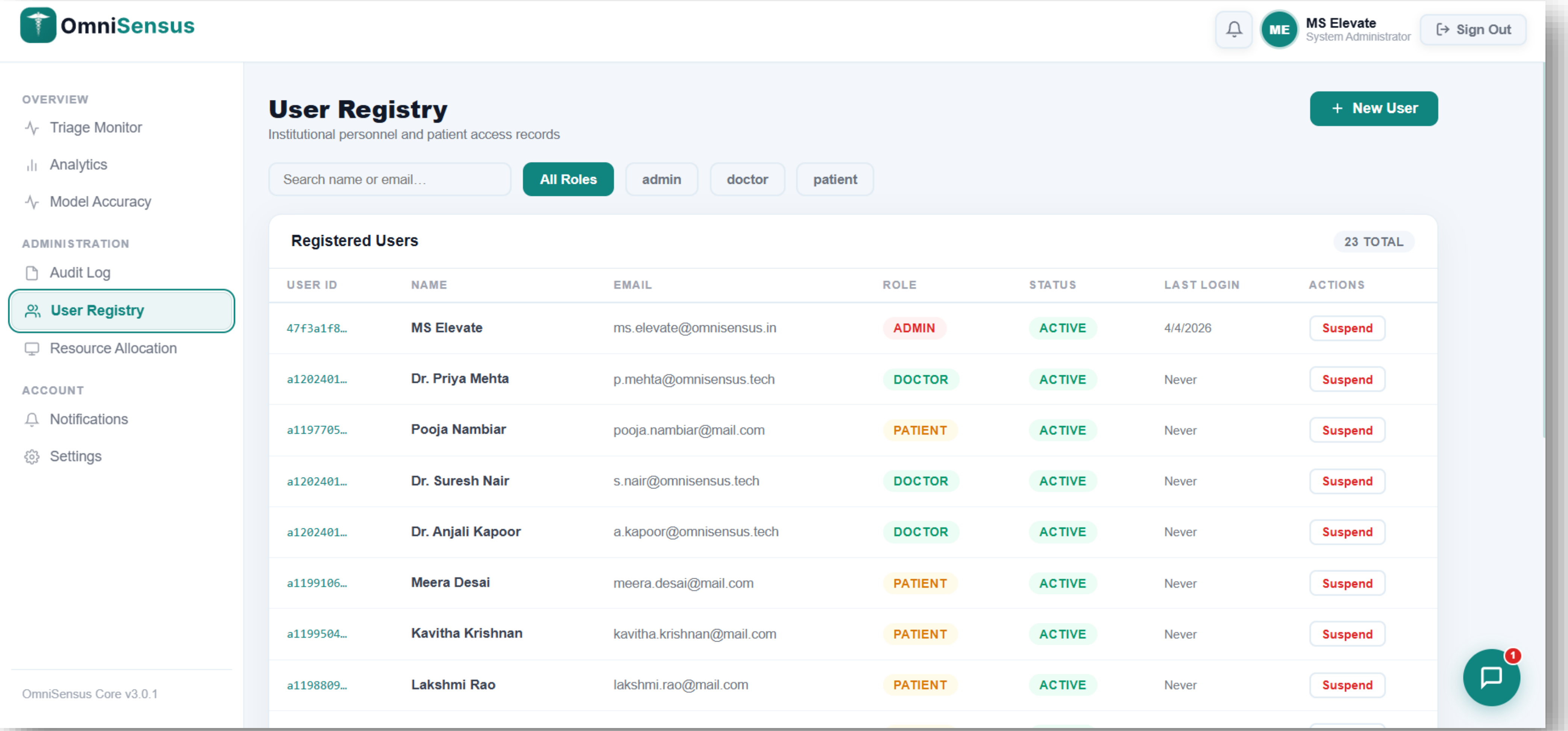
OUTPUT : ADMIN MODULE

AUDIT LOG PAGE



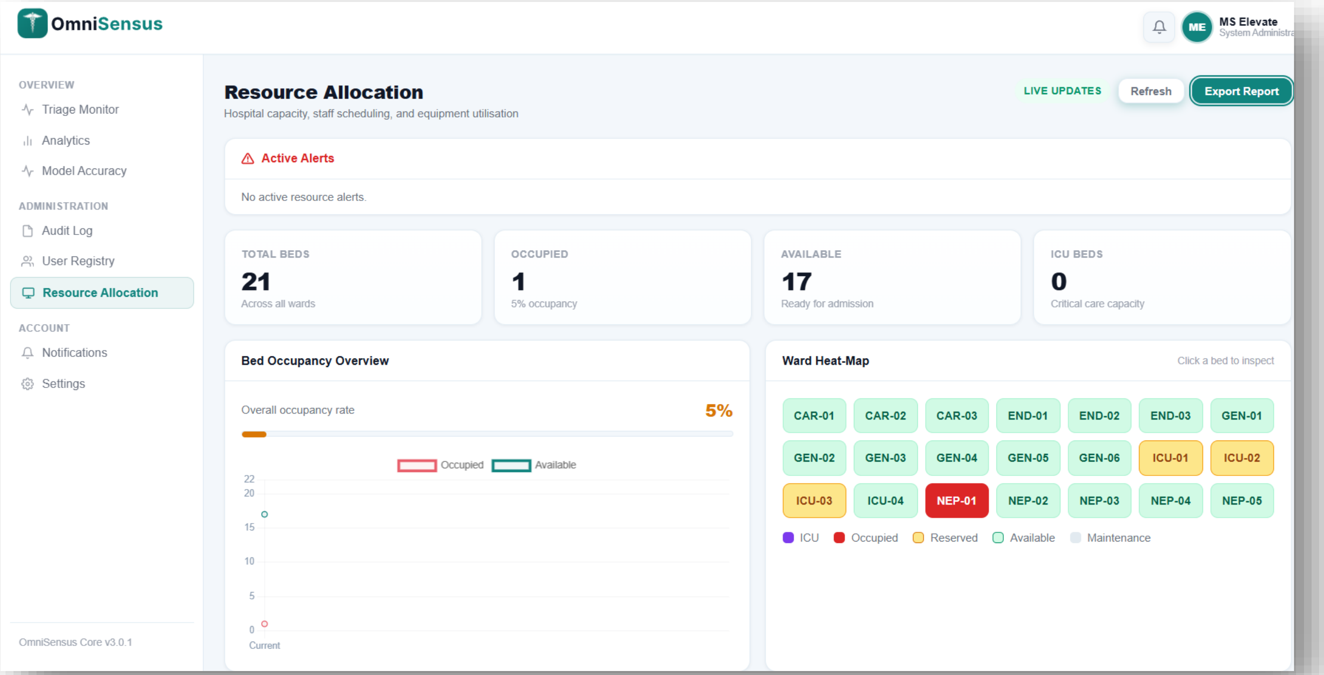
OUTPUT : ADMIN MODULE

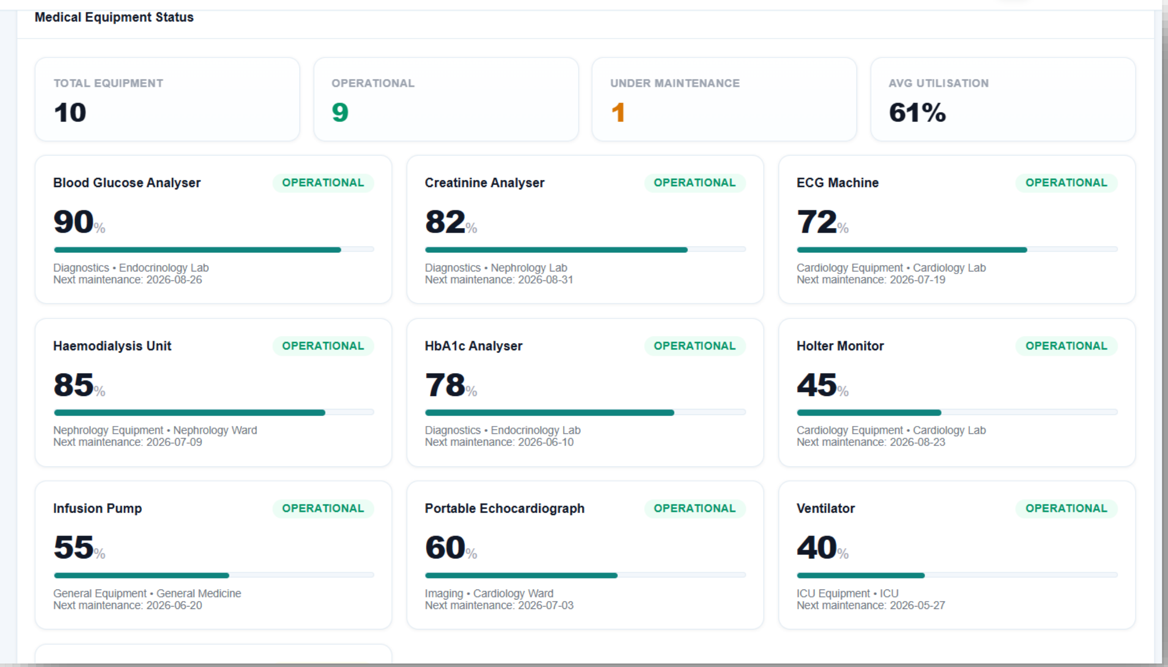
USER REGISTRY PAGE



OUTPUT : ADMIN MODULE

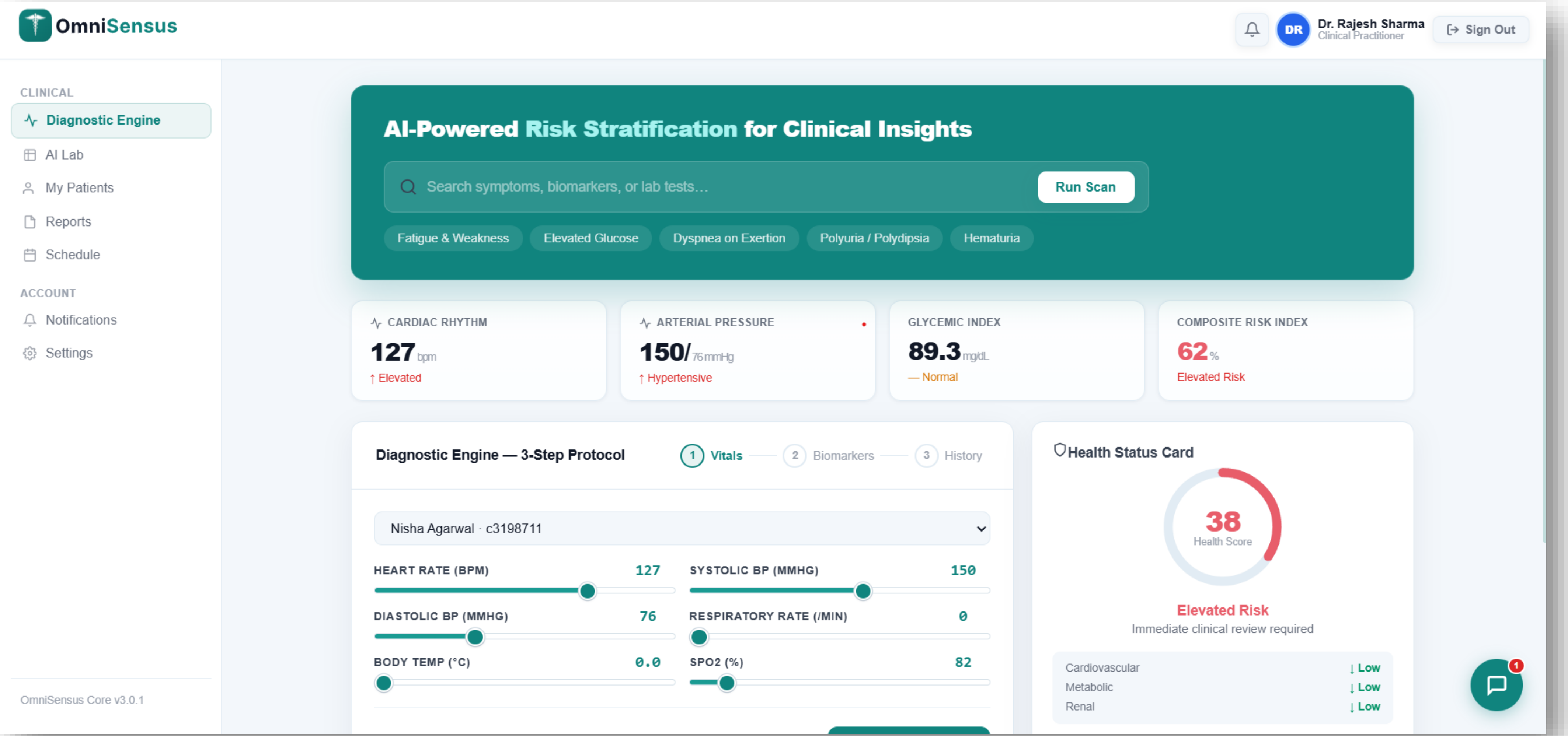
RESOURCE ALLOCATION PAGE





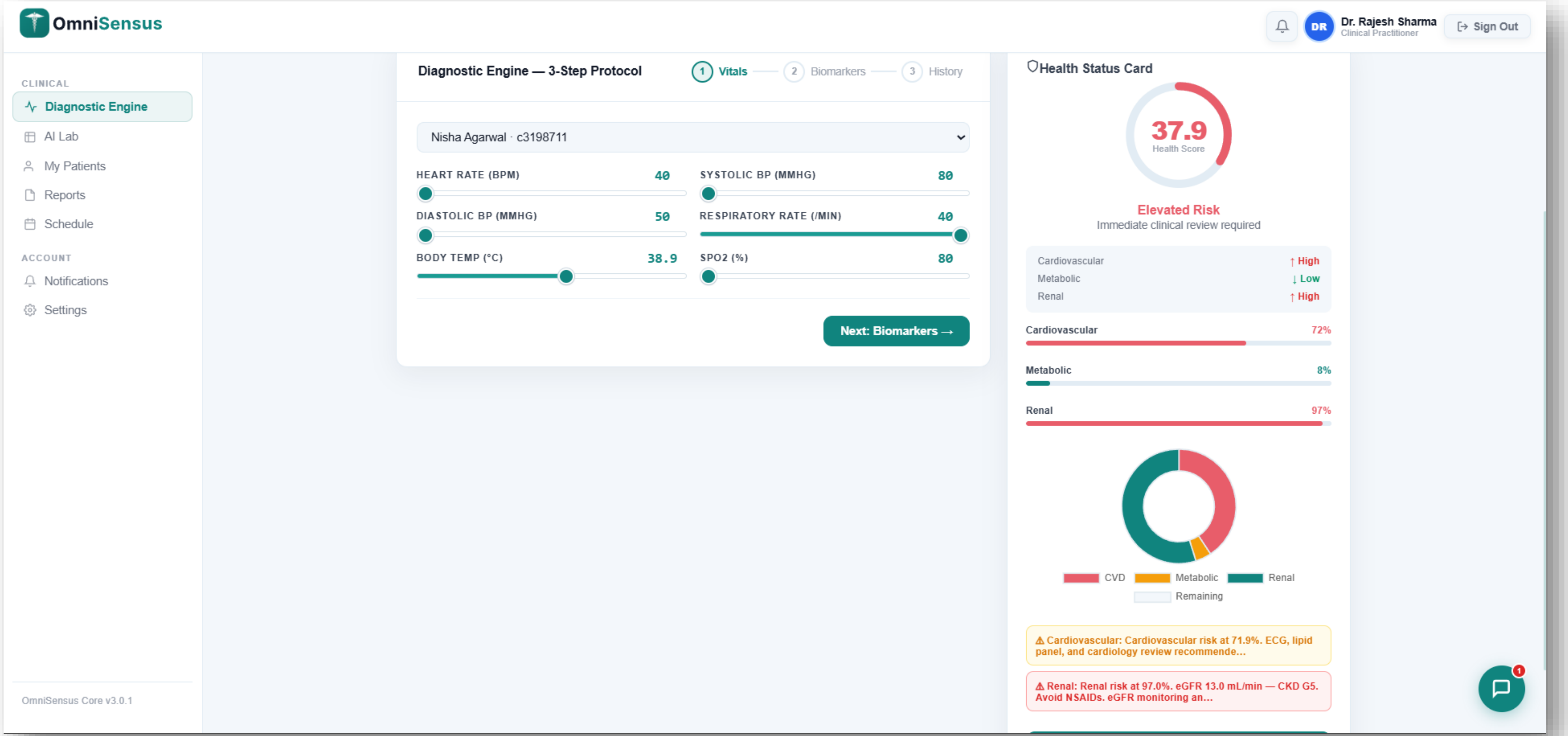
OUTPUT: DOCTOR MODULE

DIAGNOSTIC WIZARD STEP-1 (VITALS INPUT)



OUTPUT: DOCTOR MODULE

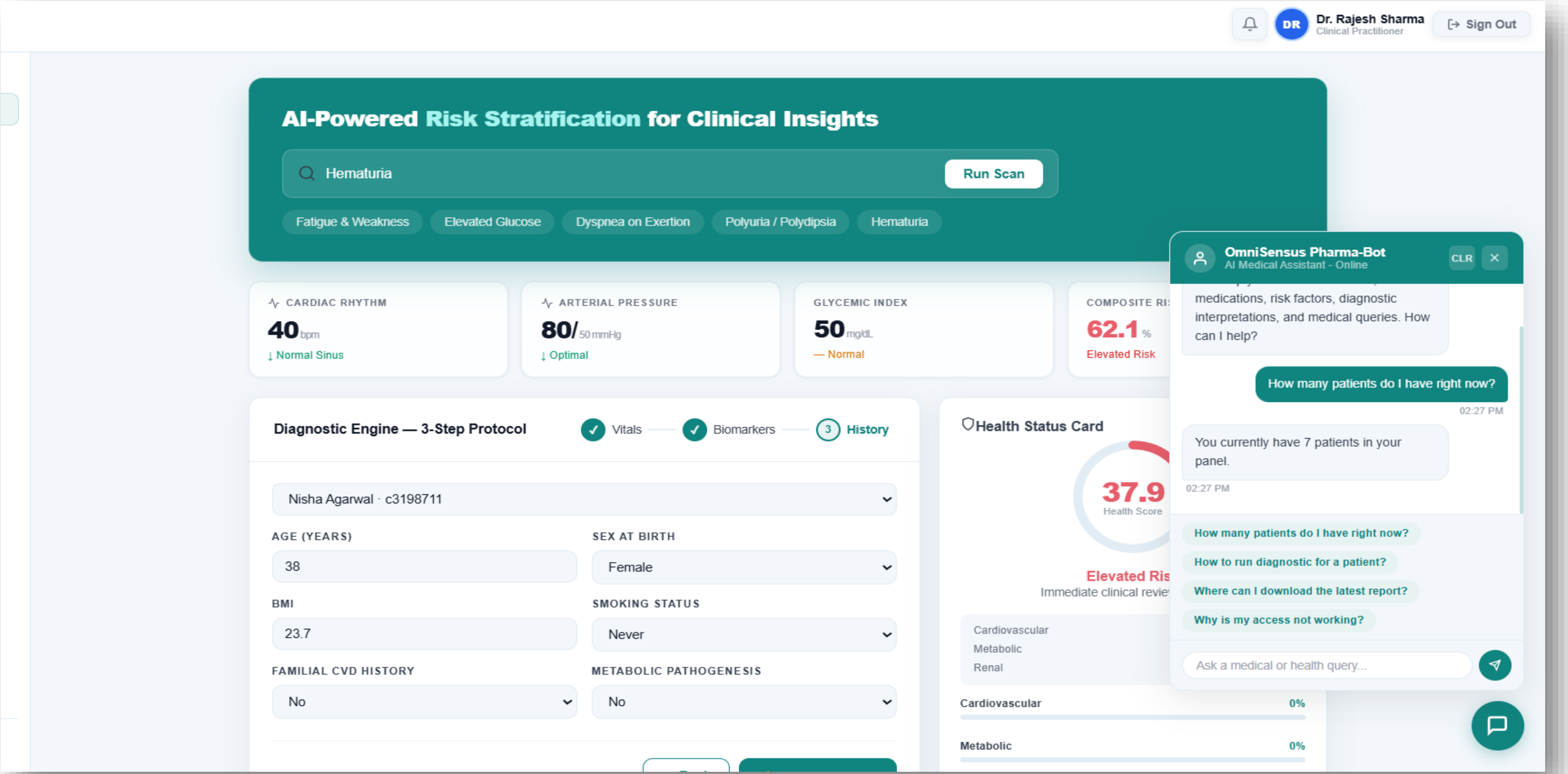
HEALTH SCORE RING + DOMAIN RISK CHART



Microsoft Elevate

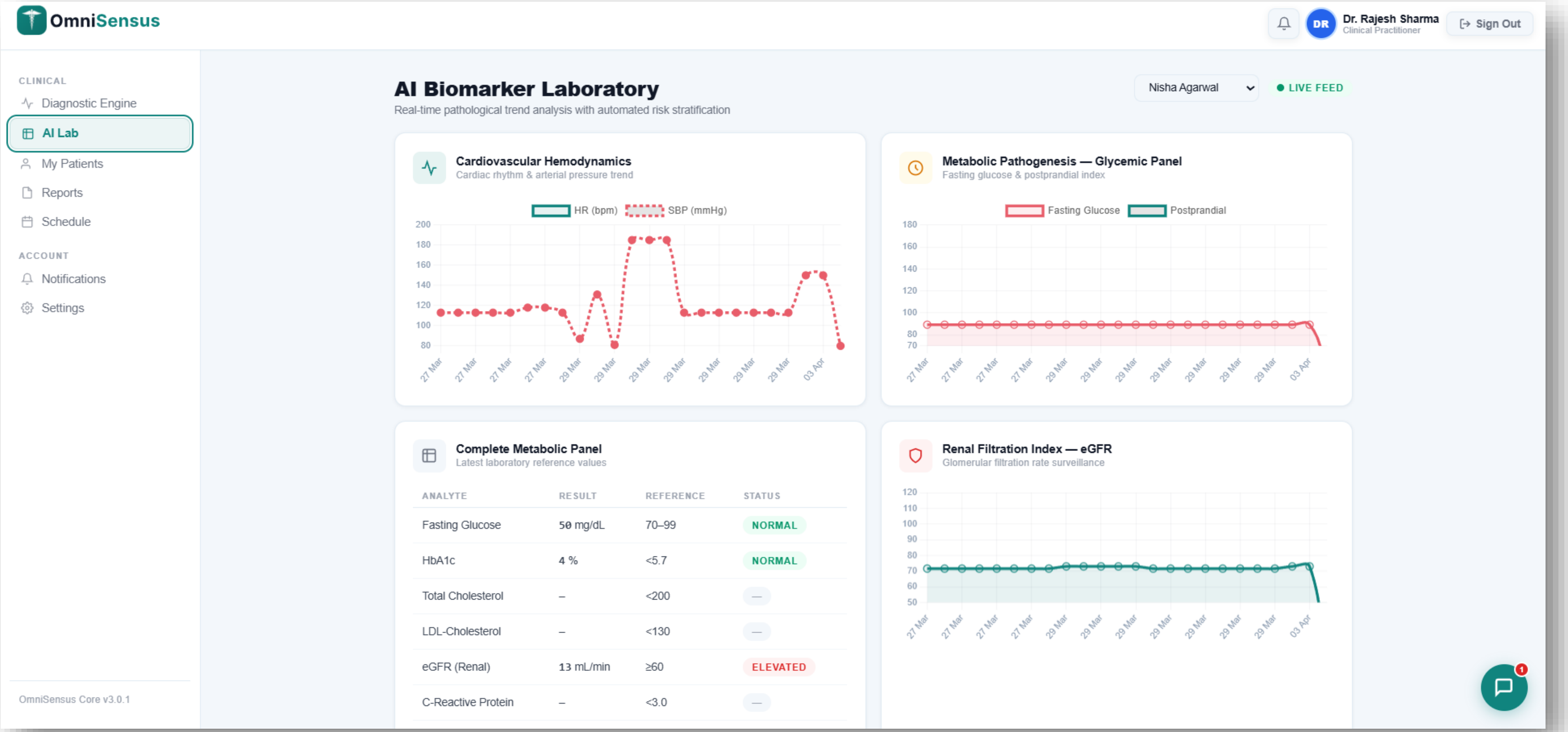
OUTPUT: DOCTOR MODULE

PHARMABOT CHAT



OUTPUT: DOCTOR MODULE

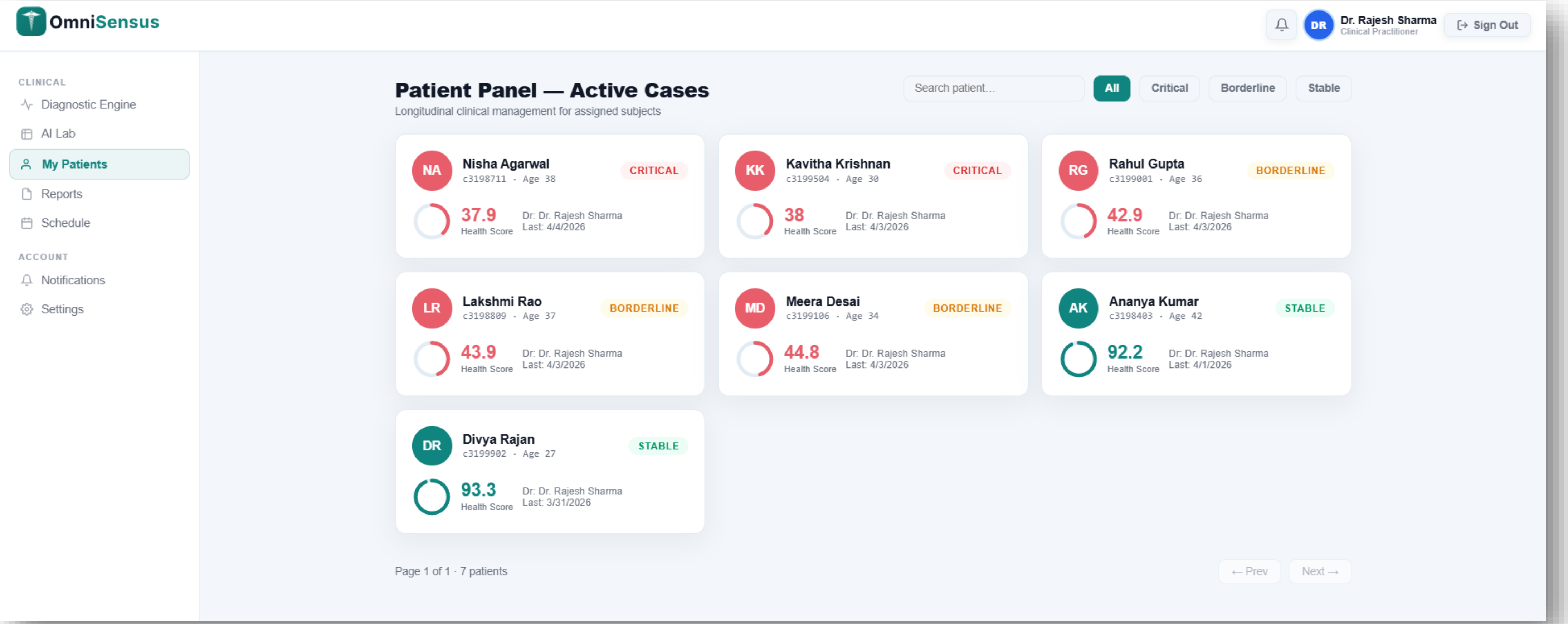
AI LAB DASHBOARD



Microsoft Elevate

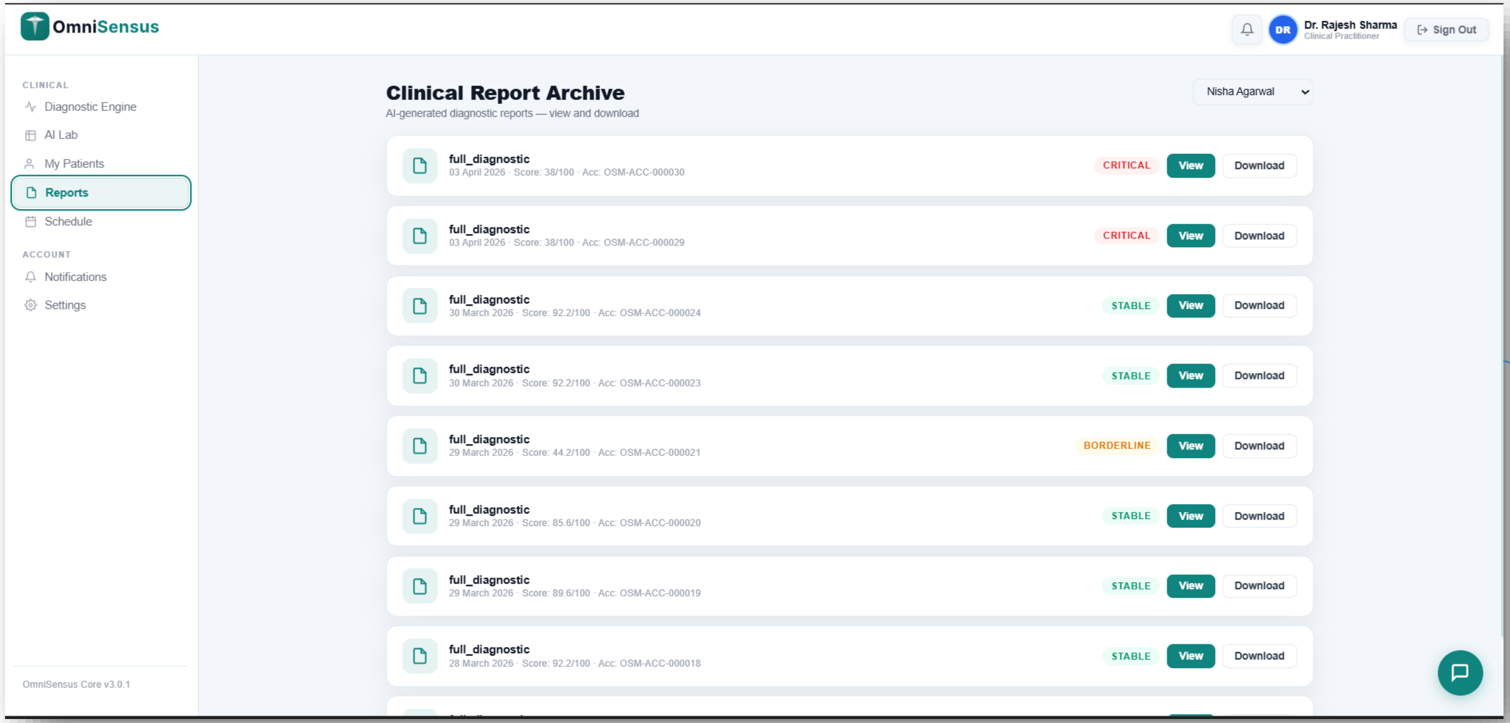
OUTPUT: DOCTOR MODULE

MY PATIENTS DASHBOARD

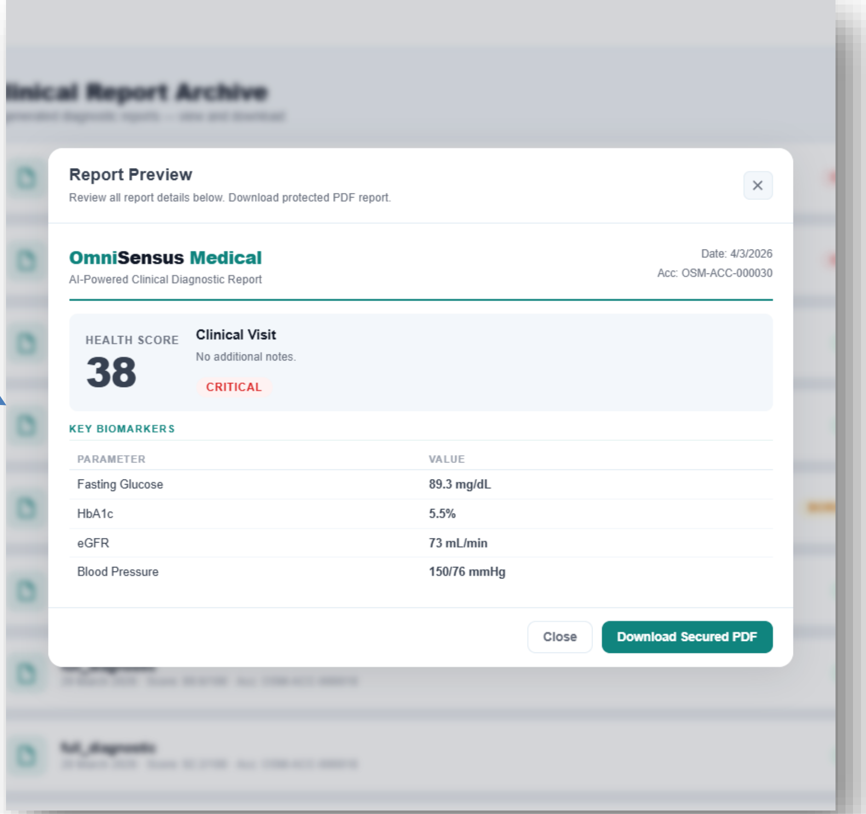


OUTPUT: DOCTOR MODULE

DOCTOR REPORTS PAGE

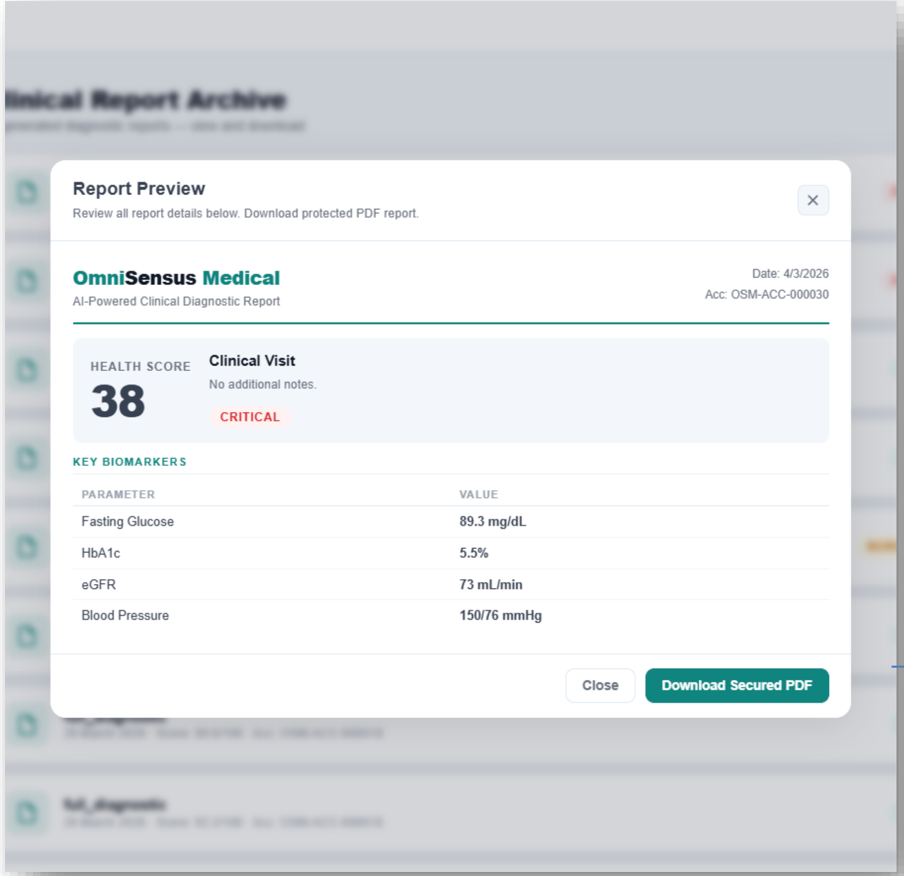


AFTER CLICK ON VIEW BUTTON



OUTPUT: DOCTOR MODULE

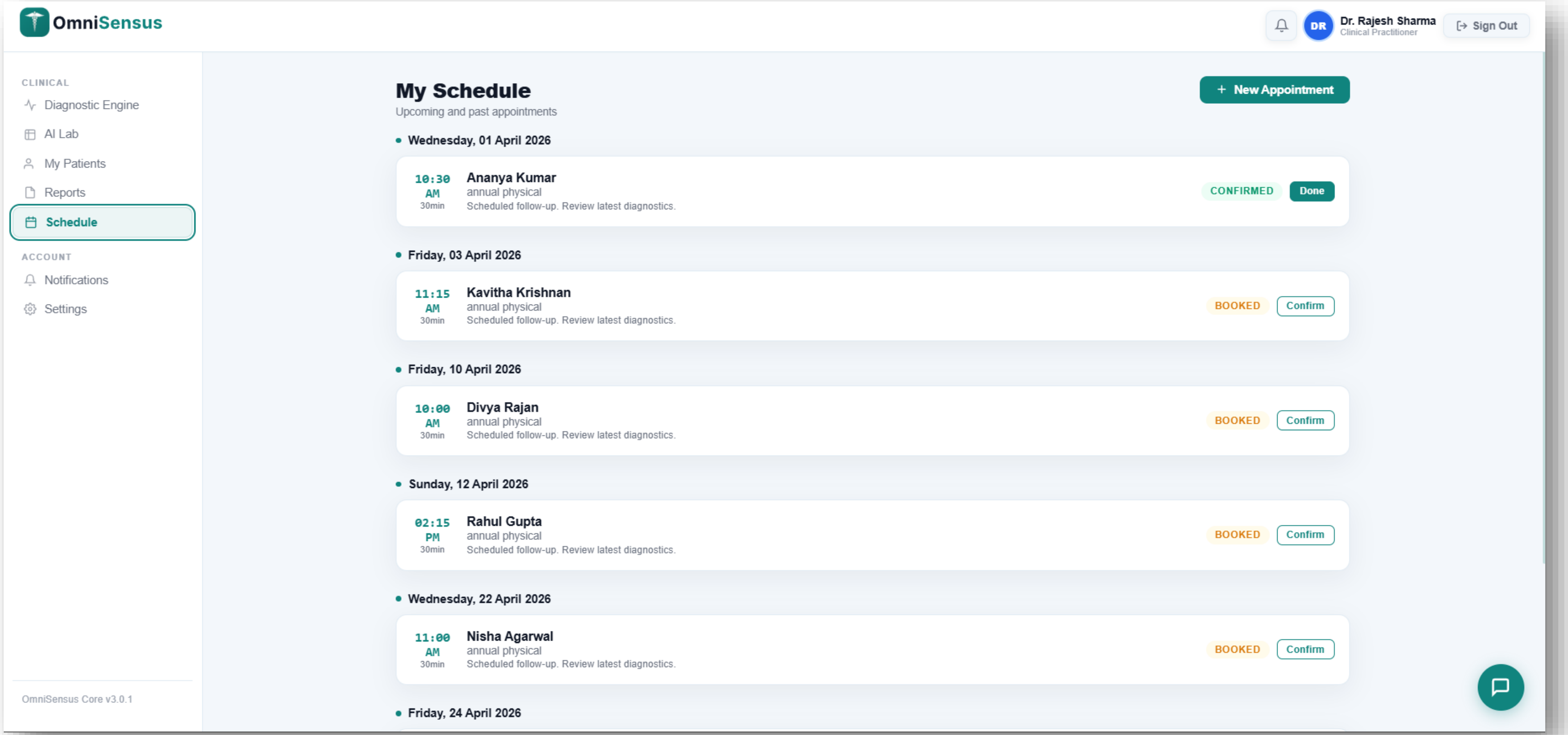
DOWNLOADED REPORTS



AFTER CLICK ON DOWNLOAD

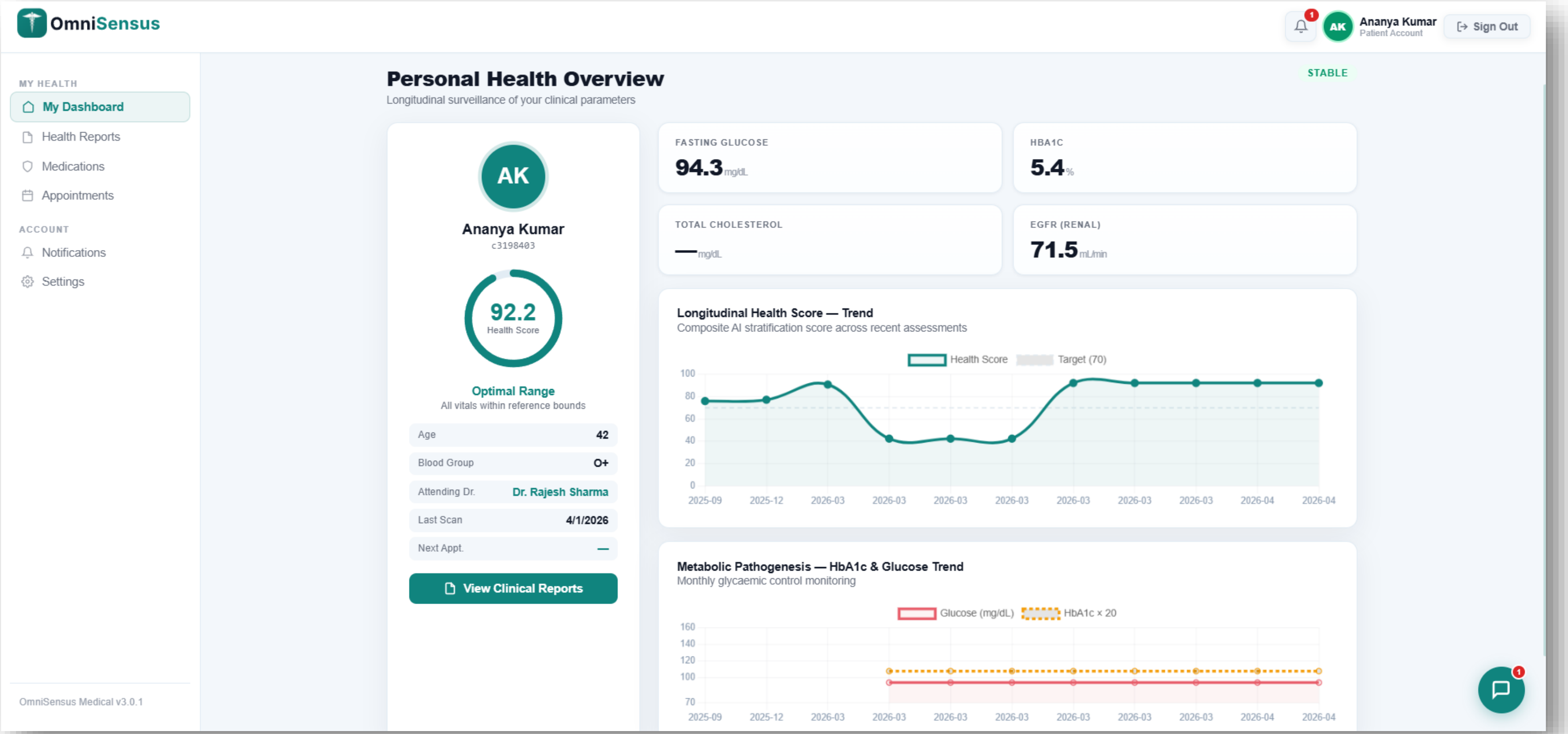
OUTPUT: DOCTOR MODULE

MY SCHEDULE PAGE



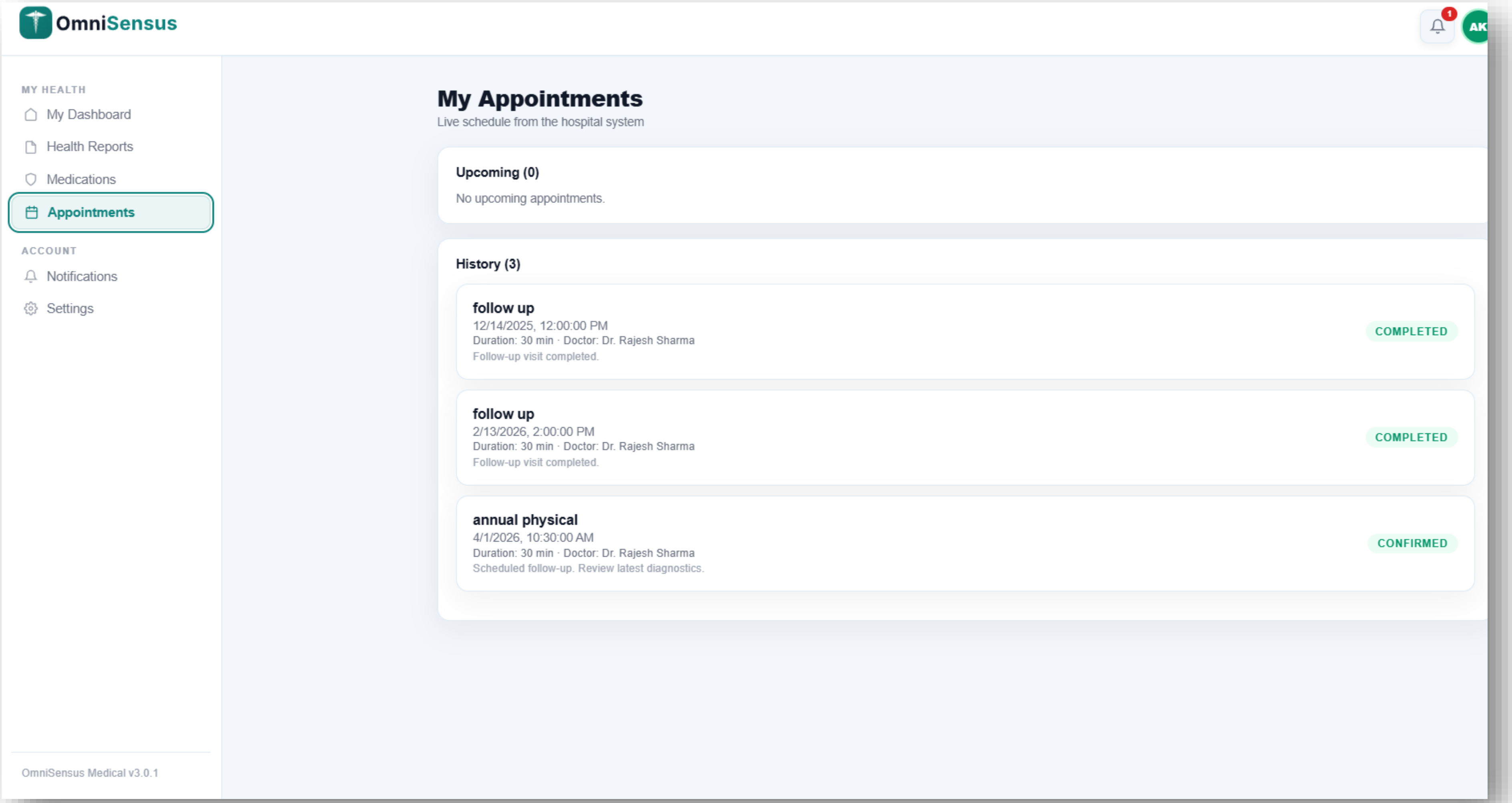
OUTPUT: PATIENT MODULE

PATIENT DASHBOARD



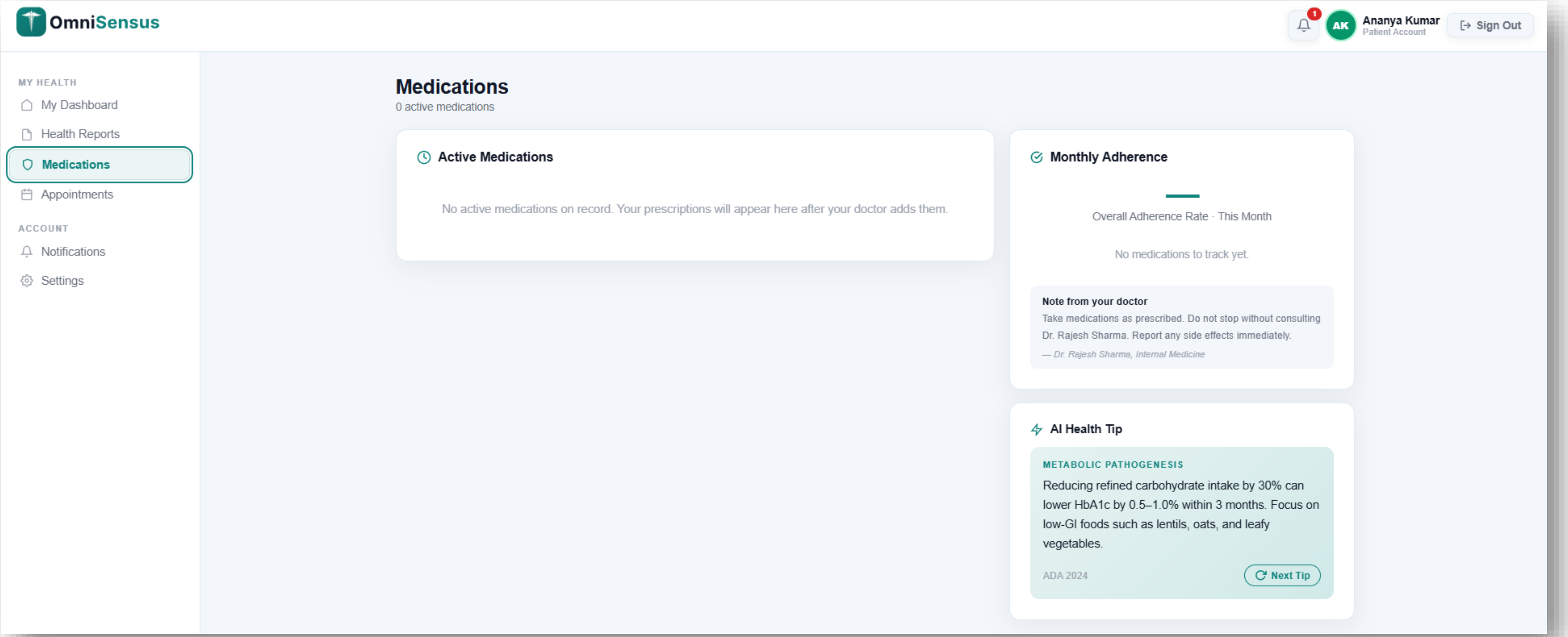
OUTPUT: PATIENT MODULE

APPOINTMENT PAGE



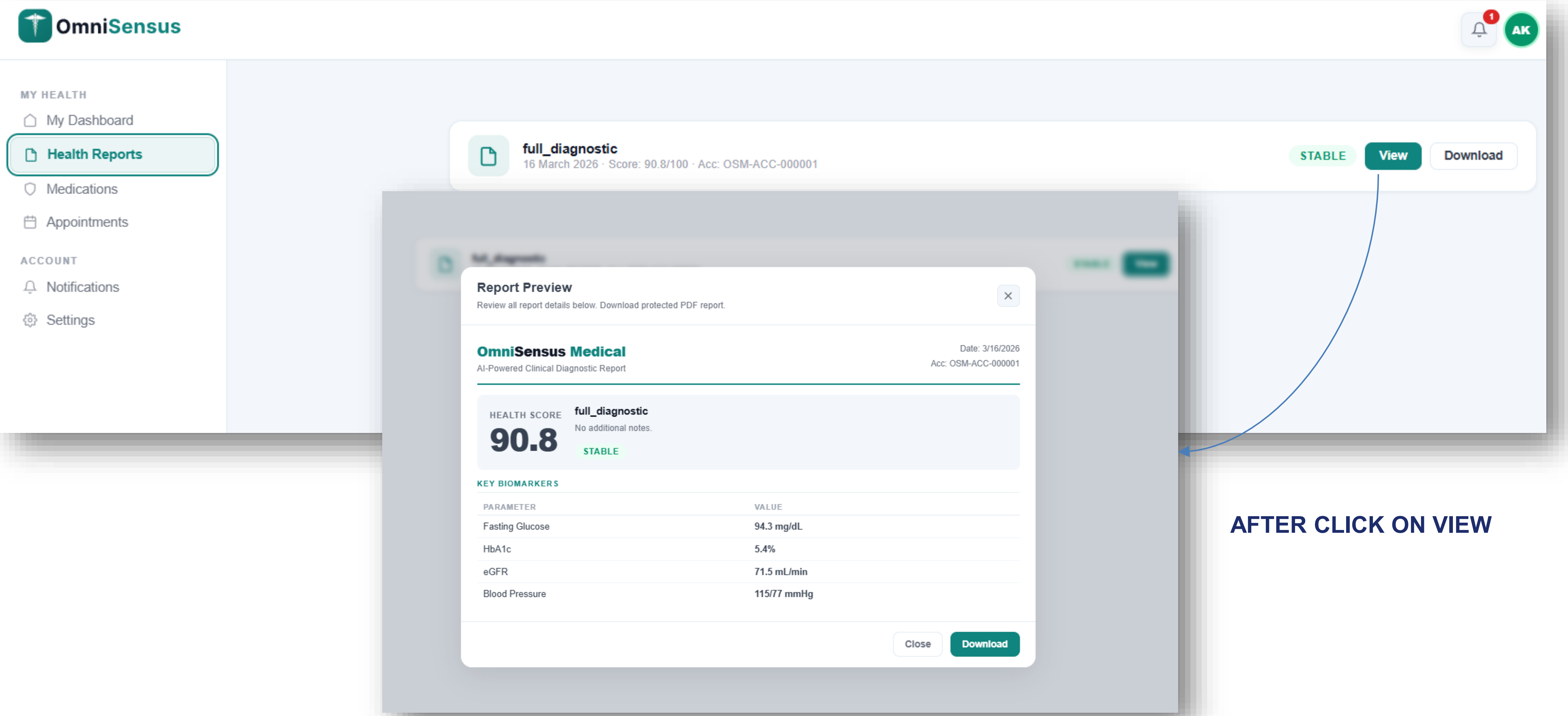
OUTPUT: PATIENT MODULE

MEDICATION PAGE



OUTPUT: PATIENT MODULE

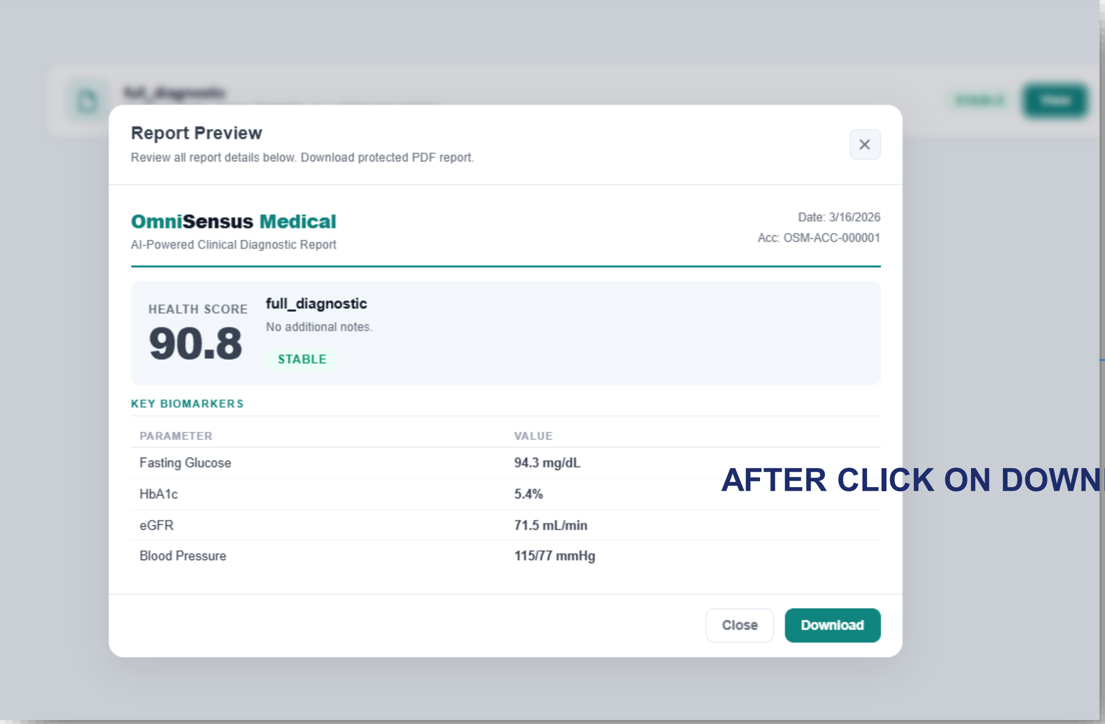
HEALTH REPORT PAGE



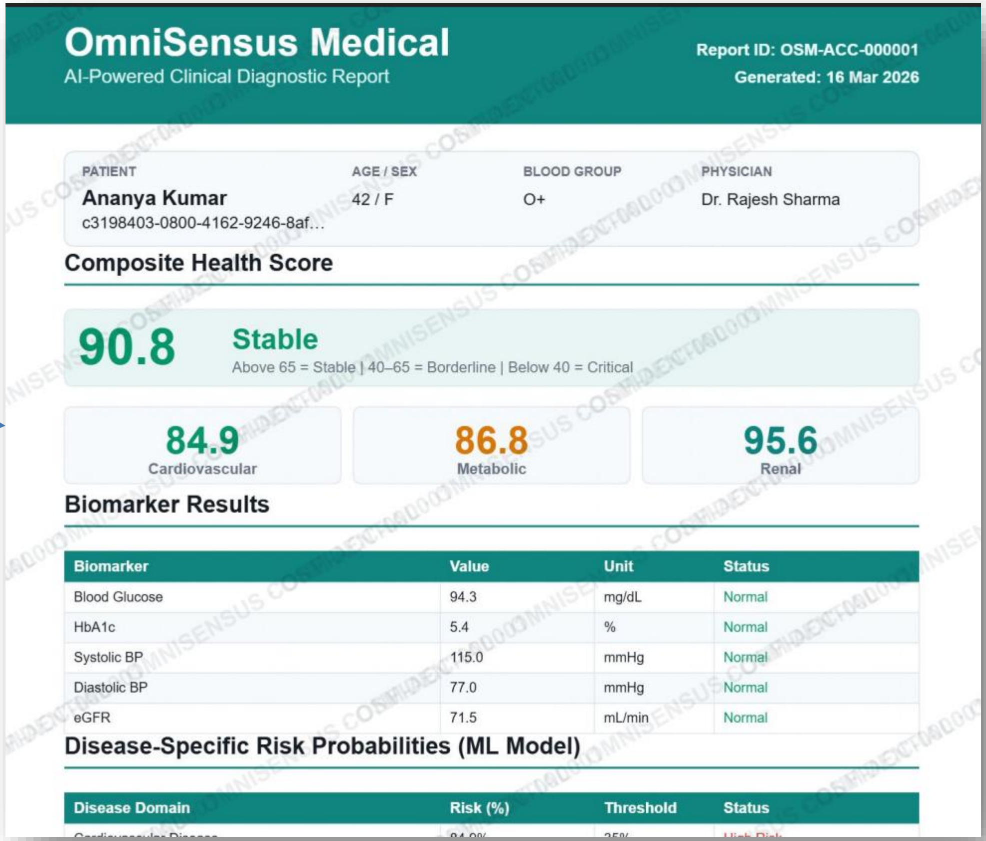
AFTER CLICK ON VIEW

OUTPUT: PATIENT MODULE

HEALTH REPORT DOWNLOADED



AFTER CLICK ON DOWNLOAD



CONCLUSION

✅ OmniSensus Medical directly addresses all 5 challenges of Problem Domain 5: Healthcare & Medical Analytics

OmniSensus Medical delivers an end-to-end, production-grade AI-powered clinical decision support platform — from biomarker input to automated PDF report generation.

The composite health scoring system (cardiovascular + metabolic + renal) gives clinicians a single, unified, interpretable wellness score — eliminating fragmented manual analysis.

SHAP-based explainability ensures every AI-generated risk score is transparent and clinically justified — building trust in AI-assisted clinical decisions.

The three-role architecture (Admin, Doctor, Patient) delivers purpose-built, secure, role-isolated workflows reflecting real hospital operational hierarchies.

All 5 Domain 5 challenges are addressed:

① Rising disease prevalence → ML early-warning system.
② Delayed diagnosis → automated 3-step wizard.
③ Resource constraints → admin triage & bed monitoring.
④ Data fragmentation → unified 27-table schema.
⑤ Predictive uncertainty → ARIMA forecasting + readmission risk model.

Platform demonstrates that production-grade ML + REST API + relational DB + modern web frontend is achievable within an academic timeline.

FUTURE SCOPE

Clinical Expansion

- Additional disease models: Liver Disease (NAFLD), Stroke Risk, Anaemia — each with validated public datasets.
- DICOM medical imaging integration (chest X-ray, ECG) for multi-modal risk combining biomarkers + imaging.
- HL7/FHIR standard compliance for interoperability with existing hospital EMR and HIS systems.

Technical Improvements

- Continuous monthly model retraining pipeline on real platform data with AUC regression gates.
- Federated learning — hospitals train locally without sharing raw patient data, preserving privacy.
- Bayesian uncertainty quantification — report P(disease) ± 95% CI, not just a point estimate.

AI & LLM Enhancements

- Upgrade Qwen 0.5B → Phi-3 Mini 3.8B on production tier for richer generative PharmaBot responses.
- Natural language vitals input — doctor types "glucose 145, BP 140/90" → LLM extracts structured data.
- LLM-generated clinical narrative summaries per run for integration into patient discharge notes.

Platform & Accessibility

- React Native mobile app — patient health monitoring, medication reminders, PharmaBot on mobile.
- Telemedicine integration — doctor-patient video consultations linked to diagnostic run results.

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[10] UCI Machine Learning Repository — CKD Dataset. Available: https://archive.ics.uci.edu

🔗 **GitHub:** https://github.com/parthvadodariya-616/omnisensus

🌐 **Live:** https://omnisensus.netlify.app/

Thank You

OmniSensus Medical — AI-Powered Clinical Intelligence

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